A seedless loop reading of Koide's relation, and dated predictions for the neutrino masses

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4 November 2026

Abstract

Koide's relation $Q = \sum m/(\sum \sqrt{m})^2 = 2/3$ holds for the charged leptons to a few parts in 10^6 and has no accepted explanation. We read it as a balance with no seed and no scale: the amplitudes $r_n = \sqrt{m_n}$ split into a shared part (the "full") and deviations that sum to zero (the "void"), and the relation is exactly $|\text{void}| = |\text{full}|$. Written as a network of relations and solved by weighted least squares, this single condition reproduces each charged lepton mass from the other two ($0.4\,\sigma$), and fixes the phase of the three amplitudes to 0.222222047, which differs from 2/9 by 1.75×10^{-7}. Applied without change to the neutrinos, with only the measured mass-squared differences as input, the loop closes only if one amplitude is in opposition and only in the normal ordering. It gives $m_{1,2,3} = 0.365,\ 8.685,\ 50.30$ meV and $\Sigma m_\nu = 59.35 \pm 0.30$ meV, hence $m_\beta = 8.84$ meV and $1.1 \le m_{\beta\beta} \le 4.1$ meV. These values were archived with a timestamp before the measurements that can refute them. A second, more speculative bet on the phase of the neutrino amplitudes is given together with its first check against JUNO.

1 Koide's relation as a balance

Koide's relation [1] concerns three masses. For three positive masses define $r_n = \sqrt{m_n}$ and split the vector $\mathbf{r}$ into its component along $(1,1,1)$ and the remainder,

$$F = 3\bar{r}^{\,2}, \qquad V = \sum_n (r_n - \bar{r})^2, \qquad \bar{r} = \tfrac{1}{3}\sum_n r_n. \tag{1}$$

The deviations $r_n - \bar{r}$ sum to zero. Since $\sum m_n = F + V$ and $(\sum r_n)^2 = 3F$,

$$Q = \frac{\sum m_n}{(\sum r_n)^2} = \frac{1}{3}\left(1 + \frac{V}{F}\right), \tag{2}$$

so $Q = 2/3$ is exactly $V = F$: the shared part and the part that cancels out carry the same weight. Equivalently $\mathbf{r}$ makes an angle of $45°$ with $(1,1,1)$, as noted by Foot [2]. In the parametrization $r_n = \mu\left(1 + k\sqrt{2}\cos(\delta + 2\pi n/3)\right)$ one has $Q = (1+k^2)/3$, so the balance is $k = 1$ and leaves the phase δ free [3, 4].

The reading we propose is a loop with no seed: no mass is a starting point, no scale enters, and the only condition is that neither part prevails. The condition fixes one ratio among three masses, not all of them.

2 Charged leptons

With the PDG 2024 masses [5], $Q = 0.6666645$ and $|Q - 2/3| = 2.2 \times 10^{-6}$, compatible with zero ($0.4\,\sigma$, the uncertainty coming almost entirely from m_τ). We solved the loop as a network of relations (nodes m_n and r_n, relations $m_n = r_n^2$ and $V = F$) with the GRM solver [9]. Given one mass, the other two remain undetermined, as they should. Given two, the third follows:

predicted	from	loop	(measured)
m_τ	$m_e,\ m_\mu$	$1776.969\,\mathrm{MeV}$	(1776.93 ± 0.09)
m_μ	$m_e,\ m_\tau$	$105.656 \pm 0.006\,\mathrm{MeV}$	(105.6583755)
m_e	$m_\mu,\ m_\tau$	$0.51087 \pm 0.00030\,\mathrm{MeV}$	(0.51099895)

Each prediction lies $0.4\,\sigma$ from the measurement. This is Koide's known success; we report it as the check that the network reproduces it.

The phase. Imposing $Q = 2/3$ exactly and using only m_e and m_μ, the phase is $\delta = 0.222222047$ (cf. Brannen [3]), against $2/9 = 0.222222222$: a difference of 1.75×10^{-7} (8 parts in 10^7). The two statements $Q = 2/3$ and $\delta = 2/9$ therefore cannot both be exact for these masses; they would predict $m_\tau = 1776.969$ and $1776.967\,\mathrm{MeV}$, indistinguishable today. Among the 95 fractions p/q with $q \le 12$ in $[0, 2\pi/3)$, the expected number within 1.75×10^{-7} of a given value is 1.6×10^{-5}. The value $2/9$ was, however, noticed after looking at these masses, and this estimate does not correct for it.

3 Neutrinos

Only the mass-squared differences are measured. With the PDG 2024 values [5] ($\Delta m^2_{21} = 7.53 \times 10^{-5}\,\mathrm{eV}^2$, $\Delta m^2_{32} = 2.455 \times 10^{-3}\,\mathrm{eV}^2$ in the normal ordering, -2.529×10^{-3} in the inverted one) the three relations

$$r_2^4 - r_1^4 = \Delta m^2_{21}, \qquad r_3^4 - r_2^4 = \Delta m^2_{32}, \qquad V = F \tag{3}$$

were solved from many starting points and for every choice of signs of the amplitudes. With all amplitudes of the same sign the loop does not close. It closes with one amplitude in opposition to the other two, as proposed for the neutrinos by Brannen [3]:

ordering	m_1 (meV)	m_2 (meV)	m_3 (meV)	Σm_ν (meV)
normal	0.365	8.685	50.30	59.35 ± 0.30
inverted	49.62	50.38	2.94	102.9 ± 0.6

The uncertainty follows from the measured differences. The inverted solution exceeds the DESI 2024 bound $\Sigma m_\nu < 72\,\mathrm{meV}$ (95 %, with CMB, ΛCDM) [6], so the loop predicts the normal ordering, with an almost massless lightest neutrino. With the mixing angles of PDG 2024 ($\sin^2\theta_{12} = 0.307 \pm 0.013$, $\sin^2\theta_{13} = 0.0219 \pm 0.0007$), the effective masses are

$$m_\beta = \Big(\sum_i |U_{ei}|^2 m_i^2\Big)^{1/2} = 8.84\ \mathrm{meV}, \qquad 1.1\ \mathrm{meV} \le m_{\beta\beta} \le 4.1\ \mathrm{meV}, \tag{4}$$

the range of $m_{\beta\beta}$ covering the two unknown Majorana phases and the uncertainties of the angles.

4 A more speculative bet: the phase on a Cantor cascade

The terms removed by the Cantor construction, 1/3, 2/9, 4/27, ..., form a geometric sequence of ratio 2/3 (the value of Q) whose sum is 1, and the charged-lepton phase is close to its second term. We fixed in advance a set of 16 candidate phases for the neutrinos (the terms t_n, their partial sums, and $2/9 + t_n$, $n \le 6$). With $Q = 2/3$, the measured ratio $\Delta m^2_{21}/\Delta m^2_{32} = 0.0307 \pm 0.0008$ allows a phase in $[0.4754, 0.4825]$ (or its mirror image); one candidate falls inside, $1/3 + 4/27 = 13/27$. With 16 candidates such a hit happens by chance about once in ten, so this is not significant by itself. Taken exactly, it fixes the ratio to 0.031230 and, with the measured Δm^2_{32},

$$\Delta m^2_{21} = (7.667 \pm 0.087) \times 10^{-5}\ \mathrm{eV}^2. \tag{5}$$

After this value was written, we compared it with the first JUNO measurement [7], $(7.50 \pm 0.12) \times 10^{-5}\,\mathrm{eV}^2$: a difference of 1.1 combined σ. JUNO's coming data will decide.

5 An earlier, rival prediction

The same author published on 30 October 2026, from a larger network of relations [10], $m_1 = 1.55\,\text{meV}$ and $\Sigma m_\nu = 60.7\,\text{meV}$ (normal ordering), with m_1 fixed by a different relation (the geometric mean of the Planck energy and the energy of the final horizon temperature). The two rules cannot hold together: in the network they are in tension by more than $40\,\sigma$. They agree on the normal ordering and on a sum close to the minimum allowed by oscillations; the mass of the lightest neutrino will tell them apart.

6 How these predictions can fail

quantity	loop	refuted if
mass ordering	normal	inverted ordering established
Σm_ν	$59.35 \pm 0.30\,\text{meV}$	settles clearly away (<55 or $>80\,\text{meV}$)
m_1	$0.365\,\text{meV}$	measured well above $0.4\,\text{meV}$
m_β	$8.84\,\text{meV}$	a signal above $\sim 15\,\text{meV}$ (today $<450\,\text{meV}$ [8])
$m_{\beta\beta}$	1.1–$4.1\,\text{meV}$	a signal clearly above $4.1\,\text{meV}$
m_τ	$1776.969\,\text{MeV}$	a measurement settles $>3\,\sigma$ away
Δm^2_{21} (Cantor)	$7.667 \pm 0.087 \times 10^{-5}$	ratio outside 0.03123 ± 0.0002

The absence of a $0\nu\beta\beta$ signal would not support the loop, since the neutrino may not be its own antiparticle.

7 Discussion

Mathematically, the loop is Koide's relation; its geometric form is Foot's, and its application to neutrinos with one amplitude in opposition is Brannen's idea. What this note adds is a reading (a balance with no seed and no scale), a derivation of the neutrino masses from the measured differences alone with no adjusted phase, an explicit statement of the tension between exact $Q = 2/3$ and exact $\delta = 2/9$, and predictions archived with a timestamp before the data that can refute them. The quark masses do not follow the relation ($Q \approx 0.85$ and 0.73 for up- and down-type, with scheme-dependent masses), and we make no claim about them. Code, tests and the dated files are available at [9].

Acknowledgements. The computations, the code and the drafting of this note were done with the assistance of an AI system (Claude, Anthropic). The ideas of the loop and of the balance between full and void are the author's.

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